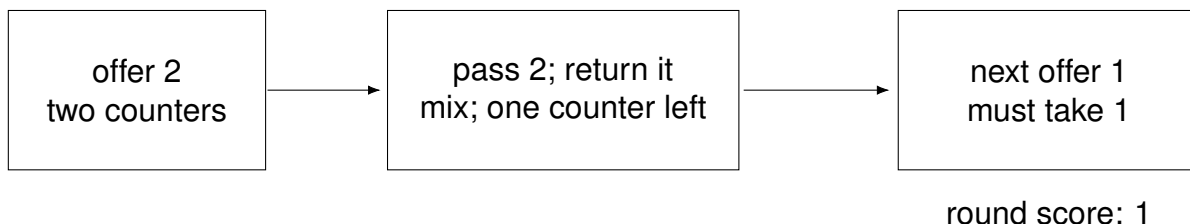


Each bag has one ticket marked 0, one marked 4 and one marked 6. Each has the same chance. Return the ticket and mix after *every* draw. The next offer is a fresh draw from the same bag.

Set the number of offers before a round. Your partner shows each offer. Take it to end the round, or pass it forever. With one counter left, you must take the offer, even if it is 0. Remove one turn counter after deciding on each offer. Only the offer you take scores.

A practice round uses different tickets: 1, 2, 5.



Problem 1: Use the 0, 4, 6 bag. Invent a taking rule for two-offer rounds and one for three-offer rounds. Keep each rule fixed while playing six rounds with a partner. Try to get a high total score. Swap roles.

Two-offer rule _____

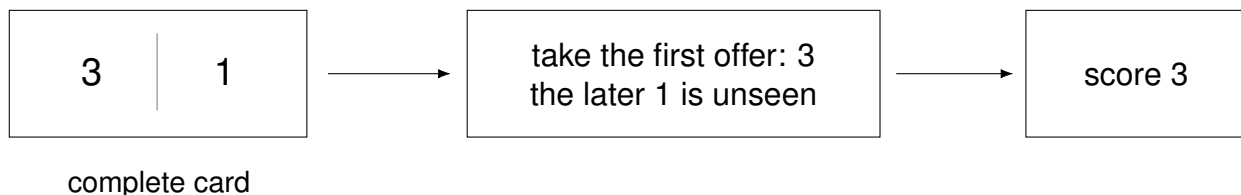
Three-offer rule _____

Round	1	2	3	4	5	6	Total
Two offers							
Three offers							

Problem 2: Can the same fixed rules give opposite winners in two six-round matches? Show possible offers for both. Can either match settle which rule has the better average over many rounds?

A card lists offers in drawing order, including offers that would remain unseen after you take one. All complete cards of the same length have the same chance.

With practice tickets 1, 3, 5, this rule takes 3 or 5 and passes 1:



Problem 3: Find first-offer choices for 0, 4 and 6 that give the highest total score on all nine cards. Use the same choices on every card. The second offer must be taken if you reach it.

0 0 score _____	0 4 score _____	0 6 score _____
4 0 score _____	4 4 score _____	4 6 score _____
6 0 score _____	6 4 score _____	6 6 score _____

First offer 0:

First offer 4:

First offer 6:

Total score on nine cards: _____

Problem 4: Which plan gives the higher total score on these 27 three-offer cards? Use every card for each plan, including cards with the same score.

A: Take 4 or 6 whenever it appears. Pass 0.

B: Take only 6 on the first offer. On the second, take 4 or 6 and pass 0.

Both plans take the last offer if they reach it.

A's total: _____ B's total: _____

0 0 0 score _____	0 0 4 score _____	0 0 6 score _____
0 4 0 score _____	0 4 4 score _____	0 4 6 score _____
0 6 0 score _____	0 6 4 score _____	0 6 6 score _____
4 0 0 score _____	4 0 4 score _____	4 0 6 score _____
4 4 0 score _____	4 4 4 score _____	4 4 6 score _____
4 6 0 score _____	4 6 4 score _____	4 6 6 score _____
6 0 0 score _____	6 0 4 score _____	6 0 6 score _____
6 4 0 score _____	6 4 4 score _____	6 4 6 score _____
6 6 0 score _____	6 6 4 score _____	6 6 6 score _____

Problem 5: The current offer is 4. Compare taking 4 with your best plan after passing 4. Which choice has the higher average score in each position? Use all possible future offers from the 0, 4, 6 bag to support your choices.

2 offers left, including this one

4

turn counters

3 offers left, including this one

4

turn counters

Choice: _____
 Average if you take: _____
 Best average if you pass: _____

Choice: _____
 Average if you take: _____
 Best average if you pass: _____

Problem 6: Find a three-offer plan for the 0, 4, 6 bag with the largest possible total on the 27 cards. Explain why no other legal plan can do better. A plan is allowed to remember earlier passed offers.

My plan:

Total on 27 cards: _____ Average score: _____

Problem 7: For each new bag, find all the first-offer choices that give the highest total score on its nine two-offer cards. Each bag has one of each shown ticket; return and mix after every draw.

Bag: 0, 3, 6

0 0 score _____	0 3 score _____	0 6 score _____
3 0 score _____	3 3 score _____	3 6 score _____
6 0 score _____	6 3 score _____	6 6 score _____

Choices: _____ Total: _____

Bag: 0, 5, 6

0 0 score _____	0 5 score _____	0 6 score _____
5 0 score _____	5 5 score _____	5 6 score _____
6 0 score _____	6 5 score _____	6 6 score _____

Choices: _____ Total: _____

Problem 8: Use a bag with tickets 0, 5, 6. Find the best taking plan for three-offer rounds and for four-offer rounds. How should the first offer 5 be treated in each? Give each plan's average score.

Offers	Taking plan	Average
Three	_____	_____
Four	_____	_____

Problem 9: The bag and draw rules stay fixed. Could adding one allowed offer lower the best possible average score? Could it leave that average unchanged? Explain your answers for any known bag of score tickets.
